

Automated structuring of research results in Data Science over papers published in relevant journals 2003-2024

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Abstract

The goal of the paper is to develop and implement a methodology for **comprehensively** structuring research in the domain of Data Science. Specifically, we download about 67 thousand research papers published in 2003-2024 in 58 relevant journals and derive 381 keywords from the current taxonomy of Data Science based on the 2012 ACM Computing Classification System. Then we describe our three-step method for **fuzzy** clustering the keywords so that those keywords that are relevant to the same papers tend to fall into the same clusters. We consider two sub periods, I:2003-2015 and II:2016-2024 and apply the method to the periods independently. This produces six clusters at each of sub periods, I and II. The found clusters are rather tight, meaningful and highly consistent between the sub periods. They correspond to six research directions such as Learning, Clustering, and Visualization. Changes in the corresponding clusters make them tighter, more up-to-date and more specific.

Keywords: data science, **taxonomy**, research topics, keyword-to-text relevance, annotated suffix tree, co-relevance, fuzzy cluster, spectral clustering, research direction

1. Introduction

Data Science as a research direction **emerged** rather recently; two **seminal** papers **marking** its birth [Cao 2017] and [Donoho, 2017] both **claim** that as about 2015. Of course, **efforts** in trying to **make sense** of data happening to get under the researcher’s eye can be traced back much earlier, especially regarding such, also rather recent, approaches, as pattern recognition, machine learning, data analysis, data mining, big data, etc. Recent reviews of Data Science developments either as a whole or of its important parts can be found in Alloghani et al. 2020 [2], Aria et al. 2020 [3], Raschka et al. 2020 [28], Martinez et al. 2021 [21], Sarker 2021 [30], Ahmed et al. 2022 [1], Shen et al. 2022 [32], Hassani and Silva 2023 [16], Oyewole and Thopil 2023 [27], Chang et al. 2024 [5], Safitra et al. 2024 [29], Sharma et al. 2025 [31]. They all express their authors’ personal views, however well informed and well balanced.

Even when the idea is to comprehensively cover his or that aspect, one would **manually** select subjects and papers to “thorough” represent that aspect, find a few hundred papers and, in the end, manually select those few to describe in greater detail (see Ahmed et al. 2022 [1], Liu et al. 2019 [19], Liu et al. 2023 [20]).

This research can also be classed as bibliometric. We attempt to structure the research in Data Science by looking at a set of relevant research papers in an automated way. It differs from the rest by the following:

- (a) It is oriented at comprehensive unbiased analysis of Data Science research **within** a period. To do that we concentrate not on specific papers but rather on specific journals. In this way, we avoid the issue of selection of relevant papers by **shifting** that to the level of journals. Specifically, we limit ourselves to journals published only by Springer Publishers because this Publisher keeps their journal contents publicly available. Being a **prime** universal publisher with no restrictions on the editors’ policies, this is representative of the entire research process. **Altogether**, we were able to identify 59 Springer journals that have relation to Data Science (see Table 1). We decided to concentrate on the period of 2003-2024 as a most recent period covering 22 years of intense

development. Some journals in our sample started publishing within that period. For these journals we take all available issues starting from the first year of publication and finishing at the last 2024 volume.

- (b) The set of keywords here is taken from a domain taxonomy, in contrast to other approaches at which keywords are defined by the authors using manual searches through dictionaries and internet publications [Ahmed et al. 2020, Aria et al. 2020, Liu et al. 2023, Sarker 2021]. Two advantages of this choice:

(b1) Comprehensiveness. It **stems** from the fact that a good taxonomy, such as the ACM CCS 2012, comprehensively covers the domain. For our Data Science Taxonomy, we have taken all the relevant branches out of the ACM CCS 2012. Our updates to the DST, however small, are based on comprehensive reviews such as (Martinez et al. 2021, Liu et al. 2019, Shen et al. 2022].

(b2) Similar **granularity** of the keywords. This follows from the fact that they are all belong to the same layer of the taxonomy.

- (c) Relevance **assessment**. Usually, to assess relevance of keywords to texts one counts the number of occurrences of keywords, possibly in a lemmatized format, in a text. We **utilize** a more sensitive and **versatile** measure based on assessment of optimal **alignments** between fragments of keywords and texts, introduced first in [Chernyak et al. 2012] by using annotated suffix tree techniques (see a computationally sound version in [Dubov 2015]). The quality of an aligned substring in this method is scored according to the conditional probability of the last symbol in the substring as a function of its prefix – a similar assessment function is used in text generating LLMs except that they involve words rather than just symbols (see Chang et al. 2024). The relevance between a keyword and text is defined as the average conditional probability over the maximal alignments [Dubov 2015, Frolov and Mirkin 2021]. Of course, this definition of relevance is computationally intensive. This is why we use abstracts, not the papers themselves: they are much shorter.
- (d) Co-relevance clusters. There can be two main directions for the analysis. One may explore the structure of associations in the set of texts: **intercitation** clusters, most influential papers/authors, etc. The other direction would be analysis of associations between keywords, aka leaf subjects in the domain taxonomy. Obviously, one should cluster together those keywords which appear together in the relevant papers. This idea **underlies** these authors' **efforts** in [Frolov et al. 2020]. However, the implementation of the idea in that paper, through inner product between text relevance vectors modified by the levels of relevance, has not been quite successful. Indeed, to follow the idea, one needs an operation that, for each pair of keywords, a and b , would compare sets T_a and T_b of texts that are relevant to, in respect, a or b . An adequate co-relevance measure should reflect the similarity between T_a and T_b . The inner product has little to do with this idea. The results of structuring the domain reported in [Frolov et al. 2020], as well as those unreported, yet tried at different periods with various journals, support this conclusion. Found clusters contribute too little to the data scatter, most clusters are not interpretable, and those interpretable ones still need a further operation of “parsimoniously lifting” to higher ranks of the taxonomy to make any sense. In this paper, we bring forward a set-to-set similarity measure which is adequate and leads to adequate clusters.

2. Our methodology

To structure Data Science research over a period, we **undertake** the following steps:

- 2.1 Download a comprehensive set of Data Science research papers.
 - 2.2 Extract a **proper** set of keywords, or categories, corresponding to the set of leaf topics of a comprehensive taxonomy of Data Science.
 - 2.3 Compute a matrix of relevance values between the keywords and the research papers.
 - 2.4 Derive a matrix of co-relevance values between the keywords according to similarity between sets of papers relevant to them.
 - 2.5 Compute fuzzy clusters of co-relevant keywords.
 - 2.6 Make conclusions on the structure of Data Science research.
- Let us describe these steps in greater detail.

2.1. Data science research publications

Considering Springer as a universal unbiased publisher and taking into account that they **recently** made all their research publications publicly available; we decided to limit ourselves to research journals published by Springer. We found 59 Springer journals relevant to the field of Data Science (see Table 1 listing them all).

Table 1. List of Springer journals in our sample.

#	Journal title	Year of start	Number of papers		
			2003-2015	2016-2024	2003-2024
1	AStA Advances in Statistical Analysis	2004	337	254	591
2	Advances in Computational Intelligence	2021	0	98	98
3	Advances in Data Analysis and Classification	2007	196	373	569
4	Annals of Data Science	2014	60	446	506
5	Annals of Mathematics and Artificial Intelligence	1990	588	452	1040
6	Annals of the Institute of Statistical Mathematics	1949	627	413	1040
7	Applied Intelligence	1991	835	4644	5479
8	Artificial Intelligence Review	1987	538	1520	2058
9	Artificial Intelligence and Law	1992	213	190	403
10	Autonomous Agents and Multi-Agent Systems	1998	380	402	782
11	Behaviormetrika	1974	133	235	368
12	Cognitive Computation	2009	343	995	1338
13	Computational Statistics	1999	695	877	1572

1 4	Computing and Visualization in Science	1997	375	65	440
1 5	Data Mining and Knowledge Discovery	1997	532	650	1182
1 6	Data Science and Engineering	2016	0	259	259
1 7	EPJ Data Science	2012	69	429	498
1 8	Environmental and Ecological Statistics	1994	448	302	750
1 9	Evolutionary Intelligence	2008	131	790	921
2 0	Extremes	1998	279	259	538
2 1	International Journal of Artificial Intelligence in Education	2013	44	346	390
2 2	International Journal of Computational Intelligence Systems	2008	541	757	1298
2 3	International Journal of Computer Vision	1987	1154	1299	2453
2 4	International Journal of Data Science and Analytics	2016	0	417	417
2 5	International Journal of Information Technology	2017	0	1790	1790
2 6	International Journal of Machine Learning and Cybernetics	2010	298	1911	2209
2 7	International Journal of Multimedia Information Retrieval	2012	87	223	310
2 8	International Journal of Speech Technology	1995	388	771	1159
2 9	International Journal on Document Analysis and Recognition (IJ DAR)	1998	308	193	501
3 0	Journal of Big Data Analytics in Transportation	2019	0	92	92
3 1	Journal of Classification	1984	285	263	548
3 2	Journal of Data, Information and Management	2019	0	115	115
3	Journal of Intelligent Information Systems	1992	449	503	952

3					
3 4	Journal of Mathematical Imaging and Vision	1992	774	602	1376
3 5	Journal of Quantitative Criminology	1985	303	311	614
3 6	Journal of Systems Science and Complexity	2006	686	1006	1692
3 7	Journal of the Korean Statistical Society	2008	419	455	874
3 8	Knowledge and Information Systems	1999	946	1364	2310
3 9	Lifetime Data Analysis	1995	403	287	690
4 0	METRON	2010	137	180	317
4 1	Machine Learning	1986	724	1082	1806
4 2	Machine Vision and Applications	1988	809	883	1692
4 3	Mathematical Methods of Statistics	2007	176	118	294
4 4	Metrika	1958	684	382	1066
4 5	Minds and Machines	1991	446	300	746
4 6	Neural Computing and Applications	1993	1875	7659	9534
4 7	Neural Processing Letters	1994	531	2041	2572
4 8	Pattern Analysis and Applications	1998	508	913	1421
4 9	Pattern Recognition and Image Analysis	2006	1052	795	1847
5 0	Progress in Artificial Intelligence	2012	67	235	302
5 1	Psychometrika	1936	663	535	1198
5 2	Quantitative Marketing and Economics	2003	196	94	290
5	Social Network Analysis and Mining	2011	244	430	674

3					
5 4	Statistical Methods & Applications	1992	399	424	823
5 5	Statistics and Computing	1991	646	969	1615
5 6	Swarm Intelligence	2007	115	103	218
5 7	TEST	1992	527	476	1003
5 8	Visualization in Engineering	2013	41	39	80
5 9	World Wide Web	1998	393	919	1312

To avoid any bias, we decided to consider all the papers published in these journals. Since we started this project in 2023, we decided to take a most recent 20-year long period 2003-2022 and then added two more years while working over the project. It should be noted that 26 journals out of the 59 just started publishing after 2003, and 6 of them, after 2015. Altogether there were 69032/67757? papers in these 59 Springer journals. We thought that 2015 was the year of, so-to-speak, **tipping point** at which both deep learning proved to be an interesting phenomenon and the term Data Science became **legitimate**. Therefore, we decided to split the entire 2003-2024 period into two subperiods, 2003-2015 and 2016-2024, so that we could look at the dynamics of research processes and see whether any difference could be observed in the research topics over the two periods.

2.2. DST Taxonomy and its leaves

We consider taxonomy of the field as a useful instrument for our goal. Since taxonomy is a comprehensive hierarchical classification of the domain, the set of topics corresponding to its leaves appears to be comprehensive. Moreover, its leaves are on the same **tier** of the hierarchy, thus, the corresponding topics have similar levels of granularity.

Our Data Science Taxonomy (DST) has been obtained from the 2012 ACM Computing Classification System [29] by cutting out branches that have nothing to do with Data Science, and by adding some leaves to better **reflect** current developments (published in Frolov et al. 2018 [10]). This DTS was used by Murtagh et al (2018) [21] for assessment of the impact by individual researchers and by Frolov et al. (2020) [12] for analyzing research tendencies in data analysis over research papers published in 17 Springer journals. In 2022, SDT was further updated by adding two branches at the second tier (2.2. Multiway data analysis and 3.5. Internet of things in Table 2).

Table 2. Higher layers of DST taxonomy based on ACM CCS 2012.

Node code	Subject
1. + 0	Theory of computation
1.1.	Theory and algorithms for application domains
2. + 0	Mathematics of computing

2.1.	Probability and statistics
2.2.*	Multiway data analysis
3. + 0	Information systems
3.1.	Data management systems
3.2.	Information systems applications
3.3.	World Wide Web
3.4.	Information retrieval
3.5*	Internet of things
4.+0	Human-centered computing
4.1.	Visualization
5.+0	Computing methodologies
5.1.	Artificial intelligence
5.2.	Machine learning

Two novel branches, 5.2.6. Deep learning and 5.1.2. Knowledge engineering, were added at the third tier, and a few subjects more, like “Big data” and “Knowledge graph” were added at lower ranks. An Excel file of the updated DST is available at Github [13].

2.3. Relevance index.

As explained above, we score relevance between a key-phrase t and text T using maximal identical substrings from t and T . The relevance score of such a substring $t=t_1t_2\dots t_n$ is defined according to statistics observed over the entire text collection. This is implemented with the so-called annotated suffix tree representing the set of texts.

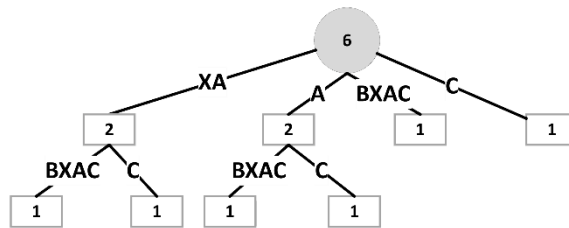


Figure 1. Annotated suffix tree for string XABXAC from Dubov 2015 [9].

Consider, for example, string XABXAC. It has six suffixes: XABXAC, ABXAC, BXAC, XAC, AC, and C. Each corresponds to a path from the root to a leaf on the tree presented in Figure 1. Edges of the path are labeled by ordered symbols from the suffix. This accords with the way the suffix tree is drawn, from the left to the right. First goes edge labeled by XABXAC, then edge labeled by ABXAC, then edge labeled by BXAC. Now is the turn of XAC, but the edge starting from X already is drawn. So XAC follows the edge corresponding to the first suffix XABXAC – till the very first different symbol, C, appears. This forces the edge to diverge. Thus, we draw a new node corresponding to the coinciding part XA, then make the node to diverge with two edges, one labeled by BXAC, the other, by C. Suffix AC generates a similar divergence over edge ABXAC: its beginning forms the first edge labeled by A and the corresponding node

issues two edges labeled by BXAC and C, respectively. This process can be reiterated to add as many other strings to the tree, as needed.

The numbers at the nodes **annotate** the suffix tree with counts of occurrences of the corresponding substring. They are updated while the process of tree drawing runs. The root is annotated with the number of suffixes. The annotation makes computation of the conditional probability of the last symbol in a string under the condition of its prefix a rather easy exercise: one just needs to divide the number at the corresponding node by the number at the **preceding** node on the path from the root. Say, the probability of C in substring XAC is $\frac{1}{2}$ (the branch on the left in Figure 1); the probability of C in substring AC is $\frac{1}{2}$ (the branch in the middle in Figure 1); and the probability of C standing alone is $\frac{1}{6}$ (the branch on the right in Figure 1).

The relevance between a keyword and text is defined as the average conditional probability over the maximal alignments which are not difficult to determine by using the technology of annotated suffix trees [Dubov 2015, Frolov and Mirkin 2021].

To make computations faster, we divide texts into shorter strings to consist of two or three consecutive words. To avoid trivial results, we use a collection of “trivial stop-words” that are skipped over in alignments. This collection contains words: «learning», «analysis», «data», «method», as well as some common suffixes: «es», «ing», «tion».

2.4. Co-relevance matrix

We don’t know any **decent** algorithm which would work at the original 67657-dimensional feature space straightforwardly. This is why we prefer converting the 381×67567 leaf-topic- to-text relevance data matrix X into a 381×381 co-relevance leaf topic similarity matrix A and do clustering over that latter matrix.

In our paper [Frolov et al. 2020] we used the inner product, $A = X^T W X$, modified by a diagonal matrix W of leaf topic weights. For leaf topic l , its weight w_l was defined to be proportional or inversely proportional to the number n_l of texts t whose relevance x_{lt} to l was at least 0.3, $x_{lt} > 0.3$. The threshold 0.3 was determined empirically after many tests. Unfortunately, the co-relevance index defined in this way was not much help. Even by using Laplacian transformation of matrix A , we managed to extract only three more or less meaningful clusters with FADDIS in [Frolov et al. 2020]; yet the total contribution of them to the data scatter was quite small.

For the time being, we tried that approach several times more. The method was consistently failing in all our experiments. Especially problematic was an issue related to those taxonomies at which the same leaf-subject was placed several times in different branches of the taxonomy tree. Say, topic “clustering” could go simultaneously into taxonomy branches oriented at methods, at theories, and applications. The inner product approach could not quite separate such plural co-occurrences. Even more **regretful** was the failure of that approach in discovery of meaningful clusters.

Therefore, we decided to apply non-linear thinking. One such approach **advances** into the idea of the text-to-leaf topic relevance threshold. Let us define, for each leaf topic l , the set T_l of texts that are relevant to l at our pre-specified threshold, $T_l = \{t: x_{lt} > 0.3\}$. Then we can use the **extent** of similarity between T_k and T_l as an estimate of the similarity between topics k and l . The size of the intersection T_k and T_l , $n_{kl} = |T_k \cap T_l|$, is a most straightforward characteristic of similarity between T_k and T_l : the greater the n_{kl} , the more similar T_k and T_l are. Two relative values, $n_{kl,k} = n_{kl} / |T_k|$ and $n_{kl,l} = n_{kl} / |T_l|$, characterize proportion of n_{kl} in T_k and in T_l , respectively. One may think of them as analogues to Precision and Recall indices in Machine Learning, assuming that one of T_k and T_l is predictor, while the other is target. Averaging them should produce a reasonable symmetric measure of similarity between sets T_k and T_l . These measures have been proposed in literature a while ago: the arithmetic average by [Kulczynski

1927], the geometric average by [Ochai 1957], and the harmonic average by [Dice 1945]. We have experimented with each of the three over our data. By far the best results have been obtained by using Kulczynski index, the arithmetic average of the $n_{kl,k}$ and $n_{kl,l}$:

$$a_{kl} = (n_{kl,k} + n_{kl,l})/2 = \frac{|T_k \cap T_l|}{2} \left(\frac{1}{|T_k|} + \frac{1}{|T_l|} \right) \quad (1)$$

This will be the only topic-to-topic co-relevance index further on.

This index value is always between 0 and 1, so that it is zero if the sets are disjoint, and it is unity if the sets coincide. In [Mirkin and Koonin, 2003] this index was analyzed, under a different name, in comparison to another, then more popular, Jaccard index which is defined as

$$J_{kl} = \frac{|T_k \cap T_l|}{|T_k \cup T_l|} \quad (2)$$

It was recognized that Kulczynski index has the following intuitively good properties that are absent from that of Jaccard:

(K1) Kulczynski index respects similarity by “inheritance”, between a set and its subset, whereas Jaccard index does not.

Consider, for example, a newspaper article, represented by a corresponding “bag of words” T , and its shorter version represented by “bag of words” $T' \subset T$, which keeps, say, only 10% of the original article’s words. Then Jaccard index value is $J = 0.1/1 = 10\%$, as well, whereas that of Kulczynski is $a = (1+0.1)/2 = 0.55 = 55\%$. The larger value is in a better accord with our intuition.

(K2) At equal-sized sets T and T' , Kulczynski index respects the size of their intersection $T \cap T'$, whereas Jaccard index does not.

Consider, for example, that the overlap $T \cap T'$ of equal-sized T and T' constitutes 40% of their sizes. That would lead to $J = 40/(40+60+60) = 0.25$, whereas $a = (0.4+0.4)/2 = 0.4$. The larger value better accords with our intuition again.

One more property of Kulczynski index, this time in its relation to the modified inner product above rather than Jaccard index:

(K3) Given that a taxonomy under consideration contains several occurrences of the same leaf-topic in different branches, Kulczynski similarity between all of them is equal to unity, so that any reasonable clustering method would put them all into the same cluster.

2.5. Additive Spectral Fuzzy Clustering

Fuzzy spectral clustering (for a recent review, see Ding et al. [7]) has gained significant attention in text analysis due to its ability to address high-dimensionality, semantic overlap, and non-linear boundaries in textual data. Our FADDIS clustering method (Mirkin, Nascimento [23]) applies fuzzy spectral clustering principles to an additive clustering model, uses a one-by-one clustering approach to iteratively extract clusters so that the number of clusters is reasonably chosen. FADDIS method relies on two core principles for thematic text analysis:

(a) *Additivity* – the similarity between keywords is considered as a sum of contributions from multiple underlying research directions; and

(b) *Non-Completeness* – the extracted clusters need not explain all the observed similarities, accommodating to the fact that some research topics are not involved yet in developed structures.

These aspects make FADDIS particularly well-suited for identifying fuzzy, overlapping research directions within taxonomically annotated collections.

Our additive model is this. Let $W = (w_{tt'})$ be a topic-to-topic similarity matrix between elements $t, t' \in T$, where $|T| = N$. A fuzzy cluster over T is represented by a membership vector $\mathbf{u} = (u_t)$, $0 \leq u_t \leq 1$, and an associated intensity value $\mu > 0$, a scale parameter that relates the membership values to the similarity scale.

According to our FADDIS clustering model, the product $\mu u_t u_{t'}$ is equal to the contribution of the cluster to the similarity between t and t' , so that the model can be represented by the following equation:

$$W = \sum_{k=1}^K \mu_k \mathbf{u}_k \mathbf{u}_k' + E \quad (3)$$

where K is the number of hidden clusters, μ_k intensity and $\mathbf{u}_k$ the membership vector of the k -th cluster ($k=1,2,\dots,K$). The matrix E is the residual similarity matrix unexplained by the clusters.

At each iteration, a single cluster is extracted to maximize the quantity:

$$g(\mathbf{u}) = \frac{\mathbf{u}' W \mathbf{u}}{\mathbf{u}' \mathbf{u}} \quad (4)$$

which represents the square root of the cluster contribution to the data scatter $Tr(W'W)$ [23]. According to the Rayleigh-Ritz theorem, this is maximized by $\mathbf{u}=\mathbf{z}$, where $\mathbf{z}$ is the eigenvector corresponding to the largest eigenvalue of W . To ensure non-negativity and normalization of the fuzzy membership vector $\mathbf{u}$, a projection operator is applied to normed $\mathbf{z}$ vector:

$$u_t = \begin{cases} 0, & \text{if } z_t < 0 \\ z_t, & \text{if } 0 \leq z_t \leq 1 \end{cases}$$

The corresponding μ value is defined by the equation $\mu = \mathbf{u}' W \mathbf{u}$ [23].

After extracting a cluster, the similarity matrix W is updated by subtracting the current cluster contribution according to the additive model in (3):

$$W \leftarrow W - \mu \mathbf{u} \mathbf{u}'$$

The algorithm proceeds until one of the following stop-conditions holds (at pre-specified K^* and ϵ values):

- (i) Cluster intensity $\mu \leq 0$;
- (ii) Relative cluster contribution to the data scatter is less than ϵ ;
- (iii) Maximum number of clusters K^* is reached.

A detailed account of the FADDIS algorithm, including its theoretical foundations, is provided in [23]. It is implemented in our GOT software [12].

3. Leaf topic clusters at two sub periods

We apply FADDIS to each of the two co-relevance matrices, one derived for period I (2003-2015), the other for period II (2016-2024). We set a stopping condition to the process of cluster extraction one-by-one over the cluster contribution to the data scatter: the process stops when cluster contribution

becomes smaller than 0.5% of the data scatter. Both periods yield 6 clusters in this way. Table 3 presents clusters' relative contributions at each period.

Table 3. Relative contributions of found clusters to the data scatter.

Cluster Number	Relative Contribution	
	2003-2015	2016-2024
1	0.3833	0.4535
2	0.0310	0.0259
3	0.0304	0.0205
4	0.0104	0.0134
5	0.0116	0.0087
6	0.0090	0.0054

The clusters of Period I and Period II appear to be highly consistent between the periods (up to a minor difference in the order of extraction, see Table 4.) Moreover, they are mostly quite homogeneous, which leads to a single research direction to annotate each of them, see Table 4.

Table 4. Research directions associated with clusters

#	Period I	Period II	Research Direction
1	Cluster 1	Cluster 1	Information Extraction
2	Cluster 2	Cluster 3	Learning
3	Cluster 3	Cluster 2	Clustering
4	Cluster 4	Cluster 4	Data Representation
5	Cluster 5	Cluster 5	Probabilistic Approaches
6	Cluster 6	Cluster 6	Visualization

Because of high degree of coincidence between Period I and Period II clusters, the following tables (Table 5 through Table 10) present them uniformly: elements of a cluster at Period I (I-cluster) go first, along with their membership values; same elements are assigned with the corresponding II-cluster membership values if they are present in that, too, in the column on the right. Or the place remains empty if the element of I-cluster does not belong in the corresponding II-cluster. It should be noted that we take the membership value of 0.15 as the threshold: only those items whose membership values are 0,15 or greater are reported as belonging to our fuzzy clusters. This threshold has been adopted empirically.

Table 5. "Information aspect" Direction

Code	DST Taxonomy leaf subjects	I-Cluster 1	II-Cluster 1
3.4.5.4./5.1.1.2.	information extraction	0.1786	0.1822
4.1.3.4.	information visualization	0.1734	0.1716
3.1.3.2.4.	join algorithms	0.1562	
5.2.4.1.2.	q-learning		0.1687

1.1.1.14.	active learning		0.1665
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The first cluster in Table 5 is by far most contributing to the data scatter. Both “data extraction” and “data visualization” are present there through the entire period 2003-2024. Additional items show the context of these activities. In Period I that was the activity of merging different data sources, exemplified by “join algorithms”, whereas that changed for “q-learning” (that is, model-free reinforcement learning) and “active learning” (that is, selecting better data points) in Period II.

These two form the strongest presence at the next research direction, the learning (see Table 6), which collects all the learning items from the DTS taxonomy in Period I, dropping just two of them in Period II.

Table 6. “Learning” Direction

Code	DST Taxonomy leaf subjects	I-Cluster 2	II-Cluster 3
5.2.4.1.2.	q-learning	0.3638	0.3222
1.1.1.14.	active learning	0.3497	0.3083
1.1.1.11.	query learning	0.2652	0.2320
5.2.2.1.	batch learning	0.2093	0.1800
1.1.1.10.	models of learning	0.1923	0.1777
5.2.6.3.	deep reinforcement learning drl	0.1921	0.2604
1.1.1.13.2.	inverse reinforcement learning	0.1896	0.2608
5.2.2.6.	active learning settings	0.1703	
3.4.4.5./ 5.2.1.1.2.	learning to rank	0.1697	0.1502
5.2.1.1.3.	supervised learning by regression	0.1582	
1.1.1.13.5./ 5.2.1.3.4.	multi-agent reinforcement learning	0.1556	0.2557

“Clustering” research direction (see Table 7) changes its position from number 3 in Period I to number 2 in Period II, reflecting the tendency to move from the “supervised learning” context to that of “unsupervised learning”. In our view, clustering is an activity oriented at modeling generalization abilities of human brain. Our findings clearly demonstrate that its time is coming. Another feature of Table 7 is the fact that item “clustering and classification” is dropped off the list in Period II to reflect the fact that now clustering is considered an activity on its own.

Table 7. “Clustering” Direction

Code	DST Taxonomy leaf subjects	I-Cluster 3	II-Cluster 2
3.2.1.4.7.	biclustering	0.3542	0.3486
3.2.1.4.4.	additive clustering	0.3380	0.3316
5.2.4.3.1.	spectral clustering	0.3279	0.3166
3.2.1.4.3	fuzzy clustering	0.3274	0.3124
3.2.1.4.1.	massive data clustering	0.3141	0.3022
3.2.1.4.2.	consensus clustering	0.2844	0.2631
3.2.1.9.2.	trajectory clustering	0.2368	0.2114
3.2.1.4.5.	feature weight clustering	0.2206	0.2297
5.2.3.2.5.	kernel-based clustering	0.2184	0.2401
3.2.1.4.6.	conceptual clustering	0.2107	0.2243
3.2.1.7.3.	graph based conceptual clustering	0.1802	0.1733
3.4.5.8.	clustering and classification	0.1556	
2.1.5.8./			
5.2.1.2.1	cluster analysis	0.1504	0.1821

The next research direction, “Representation”, has undergone a profound change indeed (see Table 8). In Period I, it was part of the process of “knowledge discovery”. That item, as well as several others, like

“knowledge graph”, “knowledge management”, “reasoning...”, etc. have been dropped off the list, so that in Period 2, this research direction stays committed exclusively to representation of various forms of input or output data including that of “smart home”.

Table 8. “Representation” Direction

Code	DST Taxonomy leaf subjects	I-Cluster 4	II-Cluster 4
3.4.2.1.	query representation	0.2383	0.2864
5.1.3.2.2.	shape representations	0.2373	0.2966
5.2.3.3.3.2.	fuzzy representation	0.2310	0.3012
5.2.3.6.2.1.	tensor representation	0.2301	0.2979
3.4.6.5.	presentation of retrieval results	0.2170	0.2664
5.1.2.2.8.	knowledge graph	0.2062	
5.1.2.2.1.	knowledge sharing	0.2046	
5.1.2.1.	knowledge management	0.2029	
5.1.3.2.4.	hierarchical representations	0.2015	0.2926
5.1.3.2.3.	appearance and texture representations	0.1948	0.3011
3.3.1.6.	knowledge discovery	0.1693	
5.1.2.1.12	reasoning about belief and knowledge	0.1561	
3.4.7.2.4.	music retrieval	0.1512	
3.2.1.11.2.	key-phrase indexing		0.2886
3.5.1.	smart home		0.2082

The next research direction, in Table 9, relates to probabilistic aspects of data science and clearly shows the changes in that from Period I to Period II: topics in regression and nonparametric statistics become outdated and change for more versatile algorithmic developments including max marginal computation, join algorithms and expectation maximization, that have been added to parallel computation already relevant in Period I.

If one may claim that previous research directions: Learning, Clustering, Representation, and Probabilistic Modeling in Tables 6-9 cover various aspects of “information extraction” in Table 5, then the next Table 10 covers the “information visualization” direction occurred in the same Table. The only difference between Periods I and II here is that the well-deserved “treemaps” visualization tool is substituted in Period 2 by more modern and versatile “task models” tools.

Table 9. “Probabilistic Approaches” Direction

Code	DST Taxonomy leaf subjects	I-Cluster 5	II-Cluster 5
2.1.8.	distribution functions	0.2740	0.2154
2.1.2.1.	maximum likelihood estimation	0.2098	0.1787
2.1.2.2.	bayesian computation	0.2032	0.2357
5.2.3.5.1.	maximum likelihood modeling	0.1910	0.1651
2.1.7.	nonparametric statistics	0.1882	
2.1.1.8.3.	bayesian nonparametric models	0.1814	
2.1.5.3.1	robust regression	0.1649	
2.1.9.	multivariate statistics	0.1616	
2.1.2.5.1.	quantile regression	0.1614	
3.2.1.3.3.	parallel computation	0.1540	0.2050
2.1.2.6.	max marginal computation		0.1815
3.1.3.2.4.	join algorithms		0.1685
2.1.3.4.	expectation maximization		0.1521

Table 10. “Visualization” Direction

Code	DST Taxonomy leaf subjects	I-Cluster 6	II-Cluster 6
4.1.4.1.	visualization toolkits	0.3470	0.3882
4.1.2.1.	treemaps	0.3336	
4.1.3.1.	scientific visualization	0.3121	0.3403
4.1.3.3.	geographic visualization	0.3096	0.3477
3.4.3.1.	personalization	0.2901	0.3266
4.1.6.	empirical studies in visualization	0.2779	0.3005
4.1.7.	visualization design and evaluation methods	0.2574	0.2518
4.1.3.4.	information visualization	0.2078	0.2033
3.4.3.2.	task models		0.2479

Let us summarize our observations. We found six clusters within the data science research efforts through the looking glass of our 381 taxonomy leaf categories, as reflected in research papers published in about six dozen Springer journals in 2003-2024. These clusters are highly consistent between Periods 2003-2015 and 2016-2024 and correspond to six research directions listed in Table 4. The table basically summarizes our findings over the structure of Data Science research in that period. One should realize that the six directions do not cover the entire area of Data Science research. Even in terms of the co-relevance matrix, the total contribution of the six clusters to the data scatter is equal to 47.57% in Period 1, and 52.74% in Period 2: about a half of the total effort. The other half of the data scatter pertains to research directions that are yet cannot be seen through our looking glass.

Also, we notice that not only are the clusters consistent over time, but the changes in them, too. We see that, in transition from Period I to Period II, the clusters drop off some loose and outdated items, whereas they acquire more focused and versatile data analysis tools to better fit with the data. It is no wonder that the total cluster contribution to the data scatter has grown by 5% in Period II.

4. Conclusion

In this paper, we provide and implement a methodology for automated comprehensive unbiased analysis of research tendencies in Data Science. We first select a long enough period, 2003-2004, with a clear-cut time point, 2015, dividing it into two subperiods. Then we select a representative set of relevant journals (59 Springer titles in Table 1 above) and download all the papers from these journals within the period. We define a DST taxonomy of the domain by using relevant fragments from the authoritative ACM Computing Classification System as is base, supplemented by very few additions reflecting some current or more detailed developments (see Table 2 above). The DST taxonomy provides us with a comprehensive and balanced list of 381 keywords corresponding to DST leaf topics.

Then we describe our three-step method for fuzzy clustering the keywords so that those keywords that are relevant to the same texts tend to fall into the same clusters. The three steps are: (I) Computing annotated-suffix-tree based keyword-to-text relevance scores; (II) Computing Kulczynski co-relevance scores between keywords; (III) Applying one-by-one FADDIS spectral clustering to the co-relevance matrix. Although some parts of the method have been published earlier, we maintain that the novel development, part (II) “Computing Kulczynski co-relevance scores”, has dramatically increased applicability of the method and transformed it into a potentially widely applicable instrument.

Independently applied to papers published at each of the two subperiods, the method has produced six clusters at each subperiod. The clusters appear rather tight, meaningful and highly consistent between the subperiods. They show that within the variety of subjects being dynamically developed a rather stable cluster substructure does exist. This substructure embraces six research directions covering about 50% of the data scatter. These research directions may be tentatively called:

- Information Extraction
- Learning
- Clustering
- Data Representation
- Probabilistic Approaches
- Visualization

Each of these directions underwent changes in the second sub period making it tighter, more up-to-date and more specific. For example, general terms “multivariate statistics”, “nonparametric statistics” and “Bayesian nonparametric models” have left cluster “Probabilistic approaches” while more specific “join algorithms” and “expectation maximization” came in (see Table 9). The least changes occurred at cluster “Learning” (Table 6). The biggest changes occurred at cluster “Representation” which included several items related to knowledge management at sub period I – they all disappeared from the cluster, making it much tighter and more specific at sub period II (Table 8). One should expect emergence of a novel cluster related to knowledge management in the future.

To the best of our knowledge no analysis of such breadth and completeness as this one has been published so far.

Among shortcomings of this work one should indicate the fact that our DST taxonomy is somewhat outdated. For example, there is no doubt that the subject “Machine learning” currently is part of the subject “Artificial intelligence”, which is not reflected in DST: In our Table 2 these items are positioned on the same tier of the DST. We do acknowledge this and humbly wait for the time when the ACM or another authoritative body develops an updated classification of computer sciences. Our future work should include reworking over an updated DST.

Another direction for future developments should be merging this method into further development of Large Linguistic Models (LLM) [18].

\acknowledgement: BM and YG gratefully acknowledge support from the Basic Research Program of HSE University in Moscow including computational resources of HPC facilities in the university.

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